

AIM

AP.140 LEARNING PLAN

<Company Long Name>

<Subject>

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Introduction

This document contains a detailed learning plan which recommends the most appropriate approaches and content to re-skill the target groups of users. It draws on the decisions captured in the Learning Strategy for Users (in the Project Readiness Roadmap, AP.070) and the information gathered in the User Learning Needs Analysis (AP.130) to define learning objectives, content, media, paths and environmental factors. In addition, it describes recommendations for reinforcement mechanisms, reward and recognition processes, and measurement methods.

Purpose

The purpose of the User Learning Plan is to:

- Propose a learning approach which meets the needs of targeted users by customizing the learning objectives, paths, and components for those groups.
- Recommend an approach which helps users to achieve the performance goals necessary for the organization to achieve the business results anticipated from the implementation.
- Recommend a reinforcement mechanism, as well as rewards and recognition process, that facilitates the transfer of newly-acquired skills to on-the-job performance.
- Recommend valid methods to measure the effectiveness of the learning events.
- Identify and remove any barriers caused by organizational, environmental, or logistical issues.

Assumptions

The following assumptions were confirmed:

- The Learning Strategy for Users (Project Readiness Roadmap, AP.070) is complete and drives this User Learning Plan
- New roles have been identified and competencies for each role have been determined (Human Performance Support Systems, AP.110).
- The User Learning Needs Analysis (AP.130) is complete, and profiles which show the gaps between future and current skills have been created for each role.

Overview

This document includes the following components:

Component	Description
Introduction	Highlights the purpose, assumptions and overview.
Learning Objectives by Role	Lists the learning objectives for each targeted role.
Learner Profiles	Describes the mandatory, highly recommended and optional learning needs by role.

Component	Description
Learning Paths by Role	Describes the approach, content and media, and sequence for learning.
Learning Reinforcement, Rewards and Recognition Methods	Lists the methods selected to facilitate the transfer and maintenance of skills acquired.
Learning Measurement Methods	Describes the means to gauge the success of the learning project.
Material Development Recommendations	Details the approach to develop the materials required.
Resource Requirements	Gives an overview of the physical and psychological issues to address for the learning project.
Learning Administration Process	Outlines the roles and tools to administer the learning events.
Appendix A	Media Selection Flowchart
Appendix B	Electronic Performance Support Systems
Appendix C	Legal Implications for Learning Verification

Learning Objectives by Role

Job Role Summary

The following summarizes highlights of job roles:

Job Role	Skills Summary (Systems Literacy, Procedure and Business)	Number of Users	Location

Learning Objectives by Role

All learning is aimed at improving performance whether it involves providing employees with knowledge they do not have currently, skills they have yet to master, or motivating them to make better use of their abilities. Learning objectives based on the results of the User Learning Needs Analysis (AP.130) provide a clear direction for learning.

A Learning Objective describes what the learner will be able to do at the conclusion of the learning. The format for a Learning Objective includes three elements which provide a framework for the content: performance, condition, and criteria. These elements answer three questions:

- What should the learner be able to do?
- Under what conditions should the learner be able to do it?
- How well must it be done?

The following template was completed for each of the roles impacted by the systems implementation.

Role 1			
Systems Literacy	Performance (The employee will be able to do X)	Condition (Given X)	Criteria (To the standard required by the job activity)
Procedural Skills	Performance	Condition	Criteria
Business Skills	Performance	Condition	Criteria

Learner Profiles

The User Learning Needs Analysis (AP.130) gave us a starting point for creating a learning approach by identifying target groups of users and the required competencies and gaps in their knowledge, skills, and aptitudes. The User Profiles mapped out the competencies by role and defined the ending point, that is, where we want users to be when the learning is completed. The Learner Profiles build on the User Profiles and the skills gaps by defining what people need to know, or learn, or do, in order to reach those objectives. The Learner Profiles include the learning prerequisites to develop competencies that include systems literacy, the procedure and business skills as defined in the User Learning Needs Analysis (AP.130).

Because of budget and time constraints, we further segmented our recommendations by identifying those pieces of the curriculum which are critical to the success of the implementation project. Each component has been labeled as:

- Mandatory
- Strongly recommended
- Optional

Systems Literacy				
Competency Area	Average Current Skill Level	Desired Skill Level	Skill Gap	Recommendations (Mandatory, Strongly Recommended, Optional)

Business Skills				
Competency Area	Average Current Skill Level	Desired Skill Level	Skill Gap	Recommendations (Mandatory, Strongly Recommended, Optional)

Procedural Skills				
Competency Area	Average Current Skill Level	Desired Skill Level	Skill Gap	Recommendations (Mandatory, Strongly Recommended, Optional)

Learning Paths by Role

In creating the Learning Paths by Role, our goal was to determine the optimal methods for transferring the needed skills to each group of learners, given their unique needs and the constraints or the opportunities provided by the organization environment. Once again, we focused on the full range of the learning experience, that is, the pre-learning preparation, the support offered during learning, and the follow up when the learners return to the job.

Learning Approach

With a systematic approach, the results of a learning program can be greatly increased. The learning process can be optimized in a number of ways. If learning is approached as a system bigger than the critical mass event only – the actual learning – the possibility of performance improvement is greatly enhanced. The systematic approach ranges from pre-learning preparation for users, support during learning and follow-up on-the-job. This includes visible participation and upper management involvement, users involvement in the learningware development process, clarification of new competency standards following the learning events, time allocation for actual learning, appropriate physical facilities and other performance support, mentorship programs, etc.

Selecting the Media

This section outlines the media or combination of media selected to meet the users' needs. Based on sound instructional decisions, the selection of the media and/or combination most appropriate to the target groups of users included a thorough understanding of the user population.

The media we chose have the following characteristics:

- Seemed most likely to produce the expected learning outcomes
- Fit the learning needs of the target group of learners
- Mirrored real work conditions and environment
- Were cost justified
- Fit the learners' culture
- Were practical to package, implement, and update
- Fit the organization's culture and history

We also considered the following factors in making the choices:

- Number of locations
- Number of learners
- Budget
- Turn-around time
- Available resources

- Need for repetition

The decision flowchart and media selection matrices used to select the media is contained in Appendix A.

Types of Media Considered

The following media were selected:

Types of Media	Media Examples
Information (The learner is not held responsible for specific actions or measurable performance.)	Conference Call E-mail memo On-line message Newsletter
Instructor-Led Learning	Case Method Demonstration Lecture/Presentation One-on-one Coaching Workshop or Lab
Paper-Based Learning	Job Aid Desk Top Manual
Technology-Based Learning	Video-Based Learning Computer-Based Learning Web-Based Learning Audio -Based Learning Interactive Distance Learning Multimedia-Based Learning (CD-ROM/CD) Oracle Tutor-Based Learning Just-in-time Learning through an Electronic Performance Support System

Learning Preferences Considered

In defining learning paths by role with appropriate media, understanding learner preferences was critical. The following learning styles were considered to facilitate enhanced delivery and retention for users:

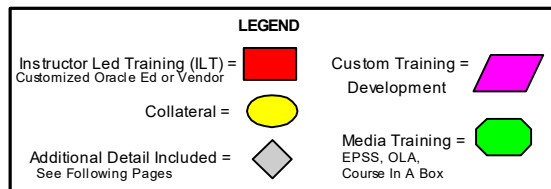
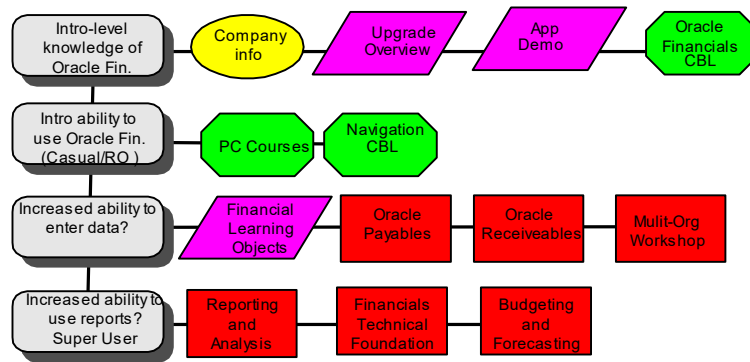
Topics	Groups	Description
Learning Styles	Audio	Audio learners absorb, retain, and recall information best when learned by hearing the information. Lecture, audio tapes, and reading methods of instruction are in this group.
	Bodily-Kinesthetic Intelligence	Kinesthetic learners absorb, retain, and recall information best when learned by manipulating objects, hands-on experiences, and trial and error.
	Visual	Visual learners absorb, retain, and recall information best when they look at graphic representations of data and concepts (graphs, pictures, models, etc.).
	Environmental Elements	Learners respond uniquely to their immediate environments when they are trying to learn something new. These elements are biological and relate to one's physical well being. These elements are sensitivities to sound, light, temperature, and design.
	Emotional Elements	Each learner has a unique emotional state that affects the learner's ability to absorb, retain, and recall information. These elements appear to be an outgrowth of both the environment and each person's emotional makeup. These elements include motivation, persistence, responsibility/conformity, and structure.
	Social Elements	Learners have preferences for working by themselves, in pairs, in teams, with colleagues, with authority, or with

Topics	Groups	Description
		groups of diverse people. These elements appear to be environmentally based.
	Physical Elements	Learners have differing physical needs related to learning. These elements are biological in origin. These elements include perceptual strengths, intake needs, time, and mobility.
	Psychological Elements	Learners have differing psychological needs related to learning. These elements appear to be both biological and environmental in origin. These elements include global/analytical thinking, hemispheric preferences, and impulsive/reflective behavior.
	Linguistic Intelligence	Linguistic learners gather information from textbooks or other literature and enjoy writing stories, essays, and reports.
	Musical Intelligence	Musical learners enjoy composing songs and ballads about subjects rather than writing reports.
	Logical-Mathematical Intelligence	Logical-mathematical learners enjoy doing research on subjects to gather the facts. They enjoy fact-based games, such as "Jeopardy" or "What's My Line."
	Spatial Intelligence	Spatial learners enjoy viewing films, pictures, charts, and maps. They use their imaginations to paint vivid pictures about subjects.
	Interpersonal Intelligence	Interpersonal learners enjoy activities that require them to interact with others. For example gathering information from interviews and enacting events, such as a meeting of King Arthur's knights of the Round Table, appeal to these learners.
	Intrapersonal Intelligence	Intrapersonal learners enjoy choosing a subject for in-depth research. They prefer to work independently to discover information.
	Analyzers	Analyzers respond to instructional strategies that include formalized lecture, data-based content, financial/technical case discussions, textbooks and bibliographies, program learning, and behavior modification.
	Organizers	Organizers respond to instructional strategies that include thorough planning, sequential order, organizational and administrative case discussions, textbooks, behavior modification, program learning, structure, and lectures.
	Personalizers	Personalizers respond to instructional strategies that include experiential opportunities, sensory movement, music, people-oriented case discussions, and group interaction.
	Visualizers	Visualizers respond to instructional strategies that include spontaneity, free flow, experiential opportunities, experimentation, playfulness, future-oriented case discussions, visual displays, aesthetics, and being involved.

Learning Paths per Role

The Learning Path combines the content, the media, and the sequence to create a flow of learning events for each targeted learning group. Learning Paths were developed with a variety of components, as illustrated below:

Accounting Path



The following shows another sample learning path for the role of a clerk in Accounts Payable.

Role/Event	Learning Content in Sequence	Delivery Media
Accounts Payable Clerk Pre-Learning Activities	Project Overview Tailored messages "What to expect" Preview/demos	Video Email Learning Lab
Learning Events	Introduction to Financials General Ledger Accounts Payable	Course-in-a-Box Instructor-led course Instructor-led course
Post-Learning Follow-up	"Using the Help Desk" Job aid design New hire learning	Lunch-and-learn session Job aid reviewer Coaching new hires

Learning Reinforcement, Rewards, and Recognition Methods

This section lists the methods selected to facilitate the transfer and maintenance of skills acquired during learning to the learner's job environment. The selection process included reviewing the learning objectives for the key roles and the associated learning paths, determining the range of options available to augment the learning path, selecting appropriate reinforcement options, and creating recommendations for each target group of learners.

Reinforcement Summary Grid

The following summarizes reinforcement measures selected for targeted groups of learners:

Targeted Groups of Learners	Job Aid	Coach/Mentor	Desk Manual	User Group	Help Desk
Role 1					
Role 2					
Role 3					
Role 4					

Learning Rewards and Recognition Process

Everyone wants to be recognized for accomplishments and rewarded for increased productivity and contribution to the success of the organization for which they work. A proven tool for motivating employee performance is recognition for a job well-done. A job well-done which is recognized provides an incentive for yet more productivity and improved performance. This plan includes guidelines to celebrate learners' accomplishments and encourage them to continue to use and strengthen their new skills. The suggested guidelines target each group of learners and specify the time frame in which they should be given. Three time frames for recognition are suggested:

1. During learning
2. At the end of learning; and
3. On the job, following learning

Guidelines for Recognition Given During Learning:

The following are guidelines for recognition measures given during learning:

- Informal
- Fun
- Opportunities for everyone to participate
- Spontaneous
- Varied

- Culturally sensitive

Guidelines for Recognition Given at the End of Learning:

The following are guidelines for recognition measures given at the end of learning:

- Formal
- Highly-public
- Matched to achievement

Guidelines for Recognition/Rewards Given on the Job, Following Learning:

The following are guidelines for recognition and rewards measures given on the job, following learning:

- Personal, that is, tailored to the individual
- Matched to achievement
- Timely
- Cumulative

Reward and Recognition Process Chart

The following summarizes the recognition measures by target group of learners:

Target Group of Learners	Suggested Recognition During Learning	Suggested Recognition at the End of Learning	Suggested Recognition/Reward following learning
Role 1			
Role 2			
Role 3			
Role 4			

Learning Measurement Methods

The general goal of developing the learning measurement method is to create a means of gauging the success of the learning project and shape recommendations for ongoing learning.

Baseline Measures

The User Profiles by Role in the User Learning Needs Analysis captured pre-learning data about the target groups of learners and thus provided the first measurement data. By comparing the baseline data with data collected after learning, it is possible to determine how much learning has occurred. Using a control group is a further step in validating measurement data. <Company Long Name> has/has not decided to use a control group.

Measurement Methods

The following measurement plan, broken out by levels of measurement, has been designed for the learning project.

- Reaction Measures
- Learning Measures
- Skills Transfer Measures
- Result Measures

In Donald L. Kirkpatrick's four levels of evaluation, Learner Verification is a Level II evaluation, measuring learning; and Performance Assessment is a Level III evaluation, measuring behavior. Learning is defined as "the extent to which participants change attitudes, improve knowledge, and/or increase skill as a result of participating in the learning event." Behavior is defined as "the extent to which change in behavior has occurred because the participant participated in the learning event. Level III, then, describes learners' behavior on the job; what learners actually do as a result of the learning event. Do they use their newly-acquired knowledge, skills, and aptitudes in the desired manner when they return to their workplaces after learning.

The type of evaluation measures a fairly exact transfer of learning to performance. Behavioral evaluation requires skillful, up-front analysis as the performance objectives are created. Evaluators must know the task in detail in order to determine if the learners know how to perform appropriately. And since the evaluation must happen at the workplace, evaluators cannot control when, or even, whether, the learners will have to use their new skills.

Measurement Matrix

The following documents the considerations per measurement level:

	I - Reaction	II - Learning	III - Skills Transfer	IV - Results
What we want to know				

	I - Reaction	II - Learning	III - Skills Transfer	IV - Results
What might be measured				
Measurement dimensions				
Sources of data				
Data-gathering methodology				
Evaluation criteria				

Selected Measures by Target Learner Group

The following table charts the measures selected by target group of learners for each level of measurement:

Target Learner Groups	Reaction Measure	Learning Measure	Skills Transfer Measures	Result Measures

Material Development Recommendations

This section details the approach to developing the materials required in order to deliver the recommended learning. Wherever possible, available courseware will be used in order to minimize the effort required. Material development requirements will vary with the type of media selected, particularly when using on-line learning, multi-media or web-based options. Resource requirements related to these technical media are provided in the following section.

Development activities will be focused on the production of:

- Quick Reference Guides
- Learning Event Overviews
- Detailed Modular Learningware including:
 - Module Objectives
 - Business Process Maps
 - Examples
 - Step-by-step instructions
 - Exercises to confirm understanding and develop skills
- Instructor Slide Packs - Originals for making reproductions
- A learning database populated with relevant data to support the execution of the learning events

The following table summarizes the development materials, together with categorizing them into those which are tailored from existing courseware and those which are to be created:

No	Materials	Amend Only	How	Source	Create
1					
2					
3					
4					
5					
6					

The estimate of materials development effort required is summarized here:

No	Materials	Vendor Man Days	<Company Long Name> Man Days	Total
1		0	0	0
2		0	0	0
3		0	0	0
4		0	0	0
5		0	0	0
6		0	0	0
	Total	0	0	0

Resource Requirements

An effectively designed learning environment supports learning. In order to create a good learning environment it is important to pay attention to both the psychological and physical environments. This section provides an overview of the learning environment issues at <Company Long Name>. These include the following:

Area	Description
Facility and sites	Finding optimal space if a requirement for each media type requires a special facility. Instructor led learning immediately comes to mind, but many delivery scenarios have a space requirement implied. For example, if video is the option selected, where will the person watch video? If one-on-one coaching is the method of choice, where will the coaching take place? Planning in advance helps make sure of the success of the option selected. If audio/built-in sound clips/shockwave files, are critical components, headphones may be required or a facility where sound can be used. This is critical when using multimedia options, for example, Oracle On-line Learning Application.
Equipment Requirements	Most learning methods require some sort of equipment. Sometimes the equipment is obvious, that is, a tape player to listen to an audio tape, but many times the equipment requirements are complex and some of them only become apparent when the learning event is about to begin. Think about such things as graphics pads, scanners, or printers, CD players, or terminal projection equipment. Computers: Assure appropriate computer workstations with sufficient space, drivers, and so on., are available with access to the appropriate applications and visual display capability. For multimedia-based products-mix, the computers must support graphics and animation or other components which support the content.
Supplies	Pencils, pens, paper, visual aid materials, workbooks, and so on. What publication issues need to be resolved?
Materials	Manuals, quick-reference cards, audio tapes, videotapes, keyboard templates, on-line exercises, etc.
Miscellaneous	Required security clearances, special transportation, parking, requirements, attendance tracking, and so on. Access to Applications must be provided for in advance of certain web-based products-mix, such as On-line Learning Application.

Learning Administration Process

This section outlines the following:

- Learning Schedule
- Materials Delivery Plan
- Roles & Responsibilities
- Learning Events Tracking Mechanism
- Learning Events Communication
- Contact Information
- Open/Closed Issues

Learning Schedule

This following table outlines the schedule for group learning at <Company Long Name> .

Date	Job Role/Function	Event Title	Prerequisites	Location

Materials Distribution Plan

Self-Guided learning involves the use of materials such as desk manuals, job aids, CBTs , and workbooks. The following table shows the plan for distributing those materials.

Learner	Location	Materials	Date Needed	Person Responsible

Roles & Responsibilities

The learning approach relies on multiple people playing multiple roles for the project to be successful. They are:

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Appendix A - Media Selection Flowchart

The Decision Flowchart

Abbreviations used in the Flow Chart:

The following lists the abbreviations used in the flow chart:

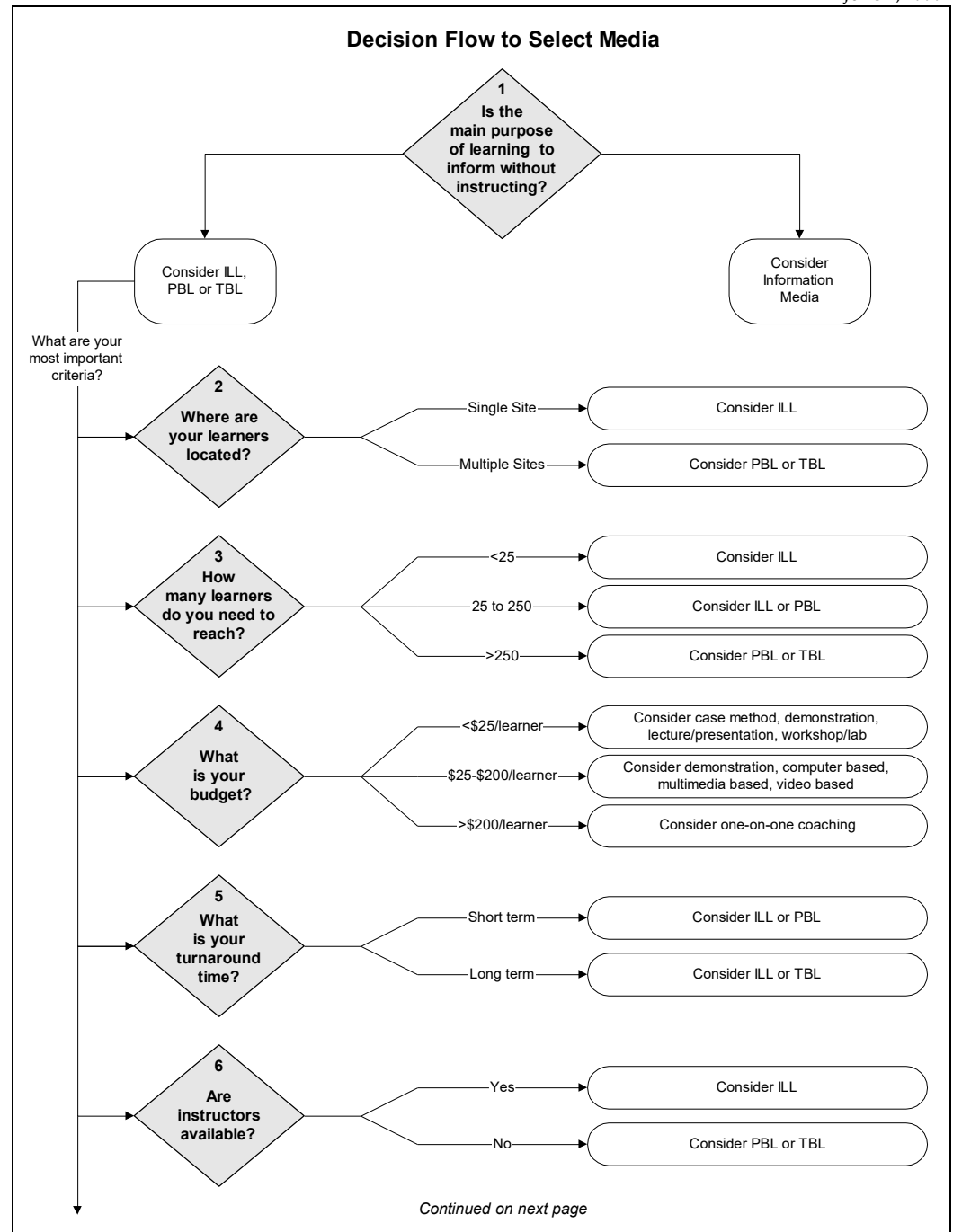
- IDL: Interactive Distance Learning
- ILL: Instructor Lead Learning
- PBL: Paper Based Learning
- TBL: Technology Based Learning

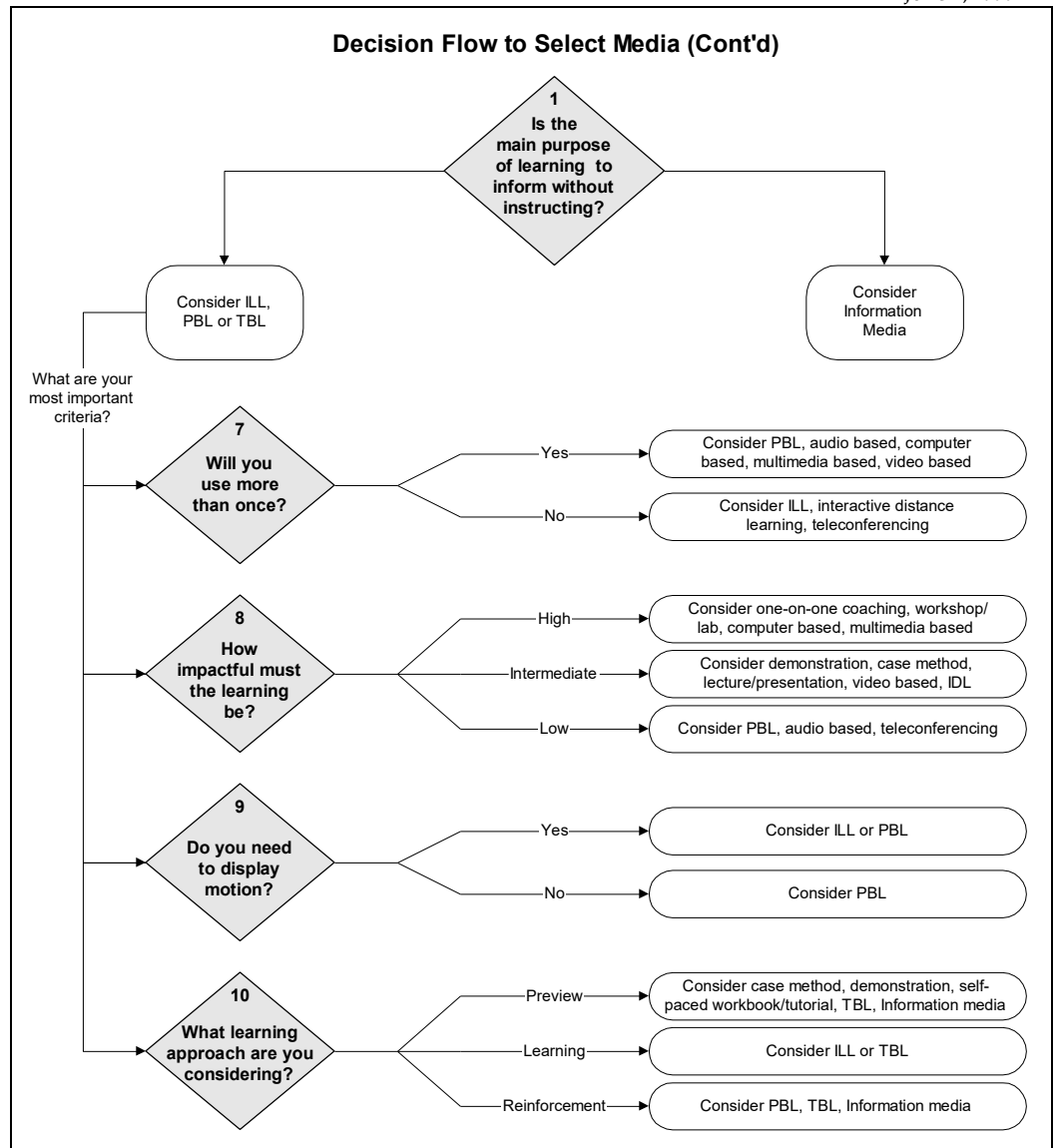
Notes on some of the Criteria in the Matrix:

The following are intended to augment the understanding of the criteria in the matrix:

- **Information:** The learner is not held responsible for measurable, specific actions or performance.
- **Instruction:** The receivers of the instruction are exposed to information and are to give demonstrable proof that they have learned.
- **Function:** The more functions the media includes, the more likely the learners are to be effective:
 - The learner is mainly exposed to the content of the change.
 - The learner can observe someone modeling the behavior(s) expected and/or has the opportunity to model the behavior(s) expected.
 - The learner has the opportunity to practice skills.
 - The learner has the latitude to repeat his/her exposure to content and/or practice of skill for better understanding, retention and mastery.
 - The learner's progress is reinforced during learning and back on-the-job.
- **Dynamic:** High interaction and participation, rationales, relevancy and on-the-job reinforcement are recommended for more in depth mastery and proficiency on-the-job.
 - Interactive level: Extent to which the learner can interact and participate within the learning media.
 - Background rationale: The content includes benefits of the change to overcome learners' potential resistance to change and support motivation to acquire the new knowledge/behaviors expected.
 - Examples: The content includes specific application relevant to the learners to make the change more concrete.

- On the job use: The media lends itself to easy use on the job as performance support until the learner achieves the required comfort, confidence, knowledge and mastery levels. On-the-job use is specially recommended in instances where a complex procedure does not need to be memorized and instances when a procedure is not often used.
- **Pace:** Learner controlled pace allows a better match to various competency levels and learning styles.
- **Intake Mode:** Multiple intake modes are recommended for longer attention spans, more in depth mastery and longer retention.
- **Feedback on Learning:** Immediate feedback is more motivational as the learner can have a better sense of progress. Timely feedback makes sure that the appropriate knowledge and skills are being acquired; if not, corrective information must be provided along with the feedback for optimal results.
 - Exercise responses, for example: the learner can complete exercises that test his/her comprehension/abilities and compare his/her responses to the Solution Key provided in the media.
 - Self-administered measurement, for example, the learner grades him/herself by assessing the number of correct responses to exercises.
 - Individualized feedback, for example, the learner gets customized feedback on knowledge or skills with additional information or practice needed by that specific learner throughout the learning process.
 - Programmed measurements, for example, by design, the learning media keeps track of the learner's results to tests and may also compare them to norms.
 - The nature of questions asked can give some indication of the level of comprehension of learners.
 - Ad hoc, for example, if the computer user makes a mistake due to the change, an error message with explanation of the change may appear on the screen.
- **Retention Rate:** High learner interaction and participation, multiple intake modes, relevancy, immediate and individualized feedback, all contribute to higher and longer term retention levels.





Learning Media Selection Matrix

The tables on the following pages provide a comparison of the following characteristics:

Audience Size	Purpose	Learning Level
Function	Dynamic	Pace
Intake Mode	Feedback on Learning	Retention Rate
Production Demand	Equipment Required	Reproducibility
Flexibility	Accessibility	Convenience of Use
Design/Development Cost	Production/Reproduction Cost	Delivery Cost
Delivery Efficiency	Communication Cost	Average Ballpark Cost

Characteristics are highlighted for each of the following learning media: The table only shows the most likely highlight(s) for each characteristic charted, that is, what media most likely lends itself to within each category of characteristics.

Class of Media	Media
Information	Conference Call Documentation E-mail memo/Bulletin Board On-line Message (for example, Help screen) Printed Memo/Newsletter
Instructor Lead Learning	Case Method Demonstration Lecture/Presentation One-on-One Coaching Workshop/Lab
Paper Based Learning	Job Aid Self-Paced Workbook/Tutorial
Technology Based Learning	Audio Based Learning Computer Based Learning Interactive Distance Learning Multimedia Based Learning (CD-ROM/CD-I) Teleconferencing Video Based Learning

Media Selection Matrix – Chart 1A: Information

The following criteria apply to information media:

	Conference Call	Documentation	E-mail memo/Bulletin Board	On-line Message (for example, Help Screen)	Printed Memo/Newsletter
Audience Size	Large group	Individual	Individual	Individual	Individual
Purpose	Information	Information	Information	Information	Information
Learning Level	Knowledge Attitude	Knowledge	Knowledge	Knowledge Thinking skills	Knowledge
Function	Exposure to content	Exposure to content Preview/Review of other media	Exposure to content	Exposure to content	Exposure to content
Dynamic	Little interaction Background rationale	No interaction Background rationale On-the-job use	No interaction Background rationale	No interaction On-the-job use	No interaction Background rationale
Pace	Pre-set by initiator	Learner controlled	Learner controlled	Pre-set by developer	Learner controlled
Intake Mode	Auditory	Visual	Visual	Visual	Visual
Feedback on Learning	None except based on questions asked	None	None	Ad-hoc	None
Retention Rate	Low level Short term	Low level Short term	Low level Short term	Low level Medium level if repeated use	Low level Short term
Production Demands	Low	Low	Low	Low	Low
Equipment Required	Telephone conferencing	Desktop publishing Printer	Computer hardware & software	Computer hardware & software	Desktop publishing Printer
Reproducibility	Not applicable	Very easy	Automatic	Automatic	Very easy
Flexibility	One time use	Reusable Easy to modify	One time use Easy to modify	Ad hoc use Easy to modify	Reusable Easy to modify
Accessibility	Through telephone Spread out use	Self-controlled Spread out use	Through computer Spread out use	Through computer Spread out use	Self-controlled Spread out use

	Conference Call	Documentation	E-mail memo/ Bulletin Board	On-line Message (for example, Help Screen)	Printed Memo/Newsletter
Convenience of Use	High: – On-the-job – At home Pre-set timing Somewhat time consuming	Very high: – On-the-job – At home Not very time consuming	Very high: – On-the-job – At home Not time consuming	Very high: – On-the-job – At home Not time consuming	Very high: – On-the-job – At home Not very time consuming
Design/Development Cost	Ratio: 2:1 (low): preparation, scheduling & coordination	Low: design, development & testing None if existing	Low: 2 hr/screen: development	Low: integration & development	Variable: low to medium development
Production/Reproduction Cost	Based on time and number of people	Low per printed master Low per unit	No cost	No cost	Low for printed master Low per unit
Delivery Cost	Callers' time Phone	Mail Readers' time	Phone Readers' time	Readers' time	Mail Readers' time
Delivery Efficiency	High for single occurrence Low for multiple occurrence	High for single site High for multiple sites	High for single site High for multiple sites	High for single site High for multiple sites	High for single site High for multiple sites
Communication Cost	Medium (long distance)	Distribution	Minimum where system exists	Minimum where system exists	Low to medium depending on quality

Media Selection Matrix – Chart 1B: Instructor Lead Learning

The following criteria apply to instructor lead media:

	Case Method	Demonstration	Lecture/ Presentation	One-on-One Coaching (On-the-job)	Workshop/Lab
Audience Size	Individual Small group	Individual Small group	Small group Large group	Individual	Small group
Purpose	Instruction	Instruction	Information	Instruction	Instruction
Learning Level	Knowledge Thinking skills	Knowledge Ability	Knowledge Attitude	Knowledge Thinking skills Ability Attitude	Knowledge Thinking skills Ability Attitude
Function	Exposure to content Practice of thinking Repetition Reinforcement	Exposure to content Observation/ Modeling Repetition/ Review of other media	Exposure to content	Exposure to content Observation/ Modeling Practice of skill Repetition Reinforcement	Exposure to content Observation/ Modeling Practice of skill Repetition Reinforcement
Dynamic	Little interaction Examples	High interaction if learner controlled Examples	No interaction Background rationale Examples	High interaction Background rationale	High interaction Background rationale Examples

	Case Method	Demonstration	Lecture/ Presentation	One-on-One Coaching (On- the-job)	Workshop/Lab
Pace	Learner controlled	Learner controlled	Lecturer controlled	Learner controlled	Balance of learner control and instructor control
Intake Mode	Visual	Tactile Visual (with motion) Auditory	Auditory Visual (if overheads or slides)	Tactile Visual (with motion) Auditory	Tactile Visual (with motion) Auditory
Feedback on Learning	Solution Key Self-administered measurement	None except based on questions asked	None	Immediate Individualized Programmed measurements	Immediate Individualized Programmed measurements
Retention Rate	Medium level Medium term	Medium level Short term	Low level Short term	High level Long term	High level Long term
Production Demands	Low	Low to medium	Low	Low	Medium
Equipment Required	Desktop publishing Printer	Actual samples or models	Slide or overhead projector	Actual samples or models	Slide or overhead projector Videotape player
Reproducibility	High with same instructor Medium with multiple instructors	High with same instructor Medium with multiple instructors	High with same lecturer Medium with multiple lecturers	High with same coach Medium with multiple coaches	High with same instructor Medium with multiple instructors
Flexibility	Non-reusable Easy to modify	Easy to modify	Easy to modify	Easy to modify	Relatively easy to modify
Accessibility	Self-controlled Spread out use	In demonstration room: on site or off site Based on availability of instructor	In lecture room: on site or off site Base on availability of lecturers	Based on availability of coaches	In workshop room: on site or off site Based on availability of workshop instructors
Convenience of Use	High: — On-the-job Somewhat time consuming	Medium Travel can be required Pre-set timing Not very time consuming	Low Travel can be required Pre-set timing Somewhat time consuming	Low Travel can be required Pre-set timing Very time consuming	Low Travel can be required Pre-set timing Very time consuming
Design/Development Cost	Ratio: 10/1 (low): design, development & testing	Ratio: 10/1 (low): design, development & testing	Ratio: 20/1 (low): design, development & testing	Ratio: 2/1 (low): design, development & testing	Ratio: 10/1 (low): design, development & testing
Production/Reproduction Cost	Low per printed master Low per unit	Based on complexity Medium per unit	Low per overhead master Low per unit	Low per printed master Low per unit	Low per overhead/slide Low per printed master High for video master Low per unit

	Case Method	Demonstration	Lecture/ Presentation	One-on-One Coaching (On- the-job)	Workshop/Lab
Delivery Cost	Instructors' time Learners' time	Travel costs Site rental Instructors' time Observers' time	Travel costs Site rental Equipment rental Lecturers' time Attendees' time	Travel costs Coaches' time Learners' time	Travel costs Site rental Equipment rental Instructors' time including Train the trainer cost if more than one instructor Attendees' time
Delivery Efficiency	High for single site Low for multiple sites	High for single site Low for multiple sites	High for single site Low for multiple sites	High for single site Low for multiple sites	High for single unit Low for multiple sites
Communication Cost	Low	Variable depending on system demonstrated	Variable depending on presentation technology	Variable depending on subject coached	Variable depending on subject of lab

Media Selection Matrix – Chart 1C: Paper Based Learning

The following criteria apply to paper based media:

	Job Aid	Self-Paced Workbook/Tutorials
Audience Size	Individual	Individual
Purpose	Instruction	Instruction
Learning Level	Knowledge Thinking skills	Knowledge Thinking skills Ability
Function	Exposure to content Preview/Review of other media	Exposure to content Practice of skill Repetition Reinforcement Preview/Review of other media
Dynamic	No interaction On-the-job use	Medium interaction Background rationale
Pace	Learner controlled	Learner controlled
Intake Mode	Visual	Visual
Feedback on Learning	None	Solution key Self-administered measurement
Retention Rate	High if repeated use	High level Medium term
Production Demands	Low	Low
Equipment Required	Desktop publishing Printer	Desktop publishing Printer
Reproducibility	Very easy	Very easy
Flexibility	Reusable Easy to modify	Easy to modify
Accessibility	Self-controlled Spread out use	Self-controlled Spread out use
Convenience of Use	Very high: – On-the-job – At home Not time consuming	Very high: – On-the-job – At home Not very time consuming

Job Aid		Self-Paced Workbook/Tutorials
Design/Development Cost	Ratio: 20/1 (low): design, development and testing	Ratio: 20/1 (low): design, development and testing
Production/Reproduction Cost	Low per printed master Low per unit	Low per printed master Low per unit
Delivery Cost	Mail Users' time	Mail Learners' time
Delivery Efficiency	High for single site High for multiple sites	High for single site High for multiple sites
Communication Cost	Low	Low

Media Selection Matrix - Chart 1D: Technology Based Learning

The following criteria apply to technology based media:

	Audio Based	Computer Based	Interactive Distance Learning	Multimedia Based (CD-ROM, CD-I)	Tele-conferencing	Video Based
Audience Size	Individual Small group	Individual	Small group Large group	Individual Small group	Small group Large group	Individual Small group Large group
Purpose	Information	Instruction	Instruction	Instruction (Combines the benefits of CBT, videotape & documentation)	Instruction	Instruction
Learning Level	Knowledge Attitude	Knowledge Thinking skills Ability Attitude	Knowledge Thinking skills Ability Attitude	Knowledge Thinking skills Ability Attitude	Knowledge Thinking skills Attitude	Knowledge Thinking skills Ability Attitude
Function	Exposure to content Observation / Modeling Practice of skill Repetition Preview / Review of other media	Exposure to content Observation / Modeling Practice of skill Repetition Reinforcement Preview / Review of other media	Exposure to content Observation / Modeling Practice of skill Repetition Reinforcement	Exposure to content Observation / Modeling Practice of skill Repetition Reinforcement Preview / Review of other media	Exposure to content Observation / Modeling	Exposure to content Observation / Modeling Practice of skill Repetition Reinforcement Preview / Review of other media
Dynamic	Little interaction Background rationale Examples	High interaction Background rationale Examples	Low to medium interaction Background rationale Examples	High interaction Background rationale Examples	High interaction Background rationale	Little interaction Background rationale Examples

	Audio Based	Computer Based	Interactive Distance Learning	Multimedia Based (CD-ROM, CD-I)	Tele-conferencing	Video Based
Pace	Pre-set by designer	Learner controlled	Instructor controlled	Learner controlled	Pre-set by initiator	Pre-set by designer Learner can stop and rewind
Intake Mode	Auditory	Visual Auditory	Visual Auditory	Visual (with motion) Auditory	Visual Auditory	Visual (with motion) Auditory
Feedback on Learning	Solution key Self-administered measurement	Immediate Individualized Programmed measurement	Solution key Programmed measurements	Immediate Individualized Programmed measurement	None except based on questions asked	Solution key Self-administered measurement
Retention Rate	Low level Short term	High level Long term	High level Long term	High level Long term	Low level Short term	High level Medium term
Production Demands	Medium to high	Very high	Low to medium	Very high	High	High
Equipment Required	Audio recording studio Audio tape player	Computer hardware & software LAN/WAN	Satellite hook up Video cameras and players	CD-I/CD-ROM playback capability	Video conferencing hardware	Video recording studio Videotape player
Reproducibility	Easy	Very easy	Easy (can be reproduced as a CD-I or CD-ROM disk)	Easy	Not applicable	Very easy
Flexibility	Reusable Hard to modify	Reusable Somewhat easy to modify	Easy to modify	Very hard to modify	Easy to modify	Reusable Hard to modify
Accessibility	Self-controlled Spread out use	Through computer Spread out use	Through links Spread out use	Through computer Spread out use	At teleconferencing site Spread out use	Through videotape player Spread out use
Convenience of Use	Very high: – On-the-job – At home Somewhat time consuming	Very high: – On-the-job – At home Very time consuming	Low: – On-the-job Somewhat time consuming	Very high: – On-the-job Very time consuming	Low Travel can be required Pre-set timing Somewhat time consuming	Very high: – On-the-job – At home Not very time consuming
Design/Development Cost	Ratio: 25/1 (medium): design, development & testing	Ratio: 150/1 (high): design, development & testing	Ratio: 50/1 (medium): development, scheduling, production	Ratio: 300/1 (high): design, development & testing	Ratio: 2/1 (low): preparation, scheduling & coordination	Ratio: 500/1 (high): design, development and testing
Production/Reproduction Cost	Medium for tape master Low per unit	High for master	High	High for master Low per unit	Based on time and number of sites	High for tape master Low per unit

	Audio Based	Computer Based	Interactive Distance Learning	Multimedia Based (CD-ROM, CD-I)	Tele-conferencing	Video Based
Delivery Cost	Mail Listeners' time	Learners' time	High per hour for studio	Learners' time	Travel costs Site rental Equipment rental Satellite time Uplink/Downlink charges Attendees' time	Mail Learners' time
Delivery Efficiency	High for single site Very high for multiple sites	Very low for single site Very high for multiple sites	Very low for single site Very high for multiple sites	Very low for single site Very high for multiple sites	Very low for single site Very high for multiple sites	High for single site High for multiple sites
Communication Cost	Low	Low	Very high: for uplink	Medium	High at this time	Low to medium

Appendix B - Electronic Performance Support Systems (EPSS)

Overview

Electronic Performance Support Systems (EPSS) typically means a computerized system designed to support the actual performance of the user at the time of need, thereby increasing productivity. EPSS allows users to perform with a minimum of support and intervention from others as users receive job task support while working and are taught “by doing.” This support is available on demand as the job is done and results in completed work.

An EPSS can contain all processes (manual and automated) of the relevant part of the organization and can prescribe the administrative procedures to be used. By using a central electronic repository, which can be linked context sensitive to the application, users have access to the administrative answer to their problem and to the way they should operate the application. In this respect, users are to a large extent autonomous. Furthermore, there is no need for them to learn every detail of the system before they actually start working with it. The introduction learning can be shorter; it will emphasize concepts and general use of the system (user interface) and teach users to find answers to actual problems, by using the EPSS.

An integrated performance support system provides a combination of specialized knowledge, searchable references and data, granular learning (cue cards and explanation), and automated tools (interest calculations, spreadsheets, templates). To develop such an integrated system requires a multidisciplinary team for proper system design, maximum efficiency, and speed. It should consist of instructional designers to establish proper objectives, functionality, and solutions, subject matter specialists for content integrity, applications architects for multimedia and graphics, and developers.

Selecting an Electronic Performance Support Systems

Following are key questions to consider when determining if EPSS is the best media:

- Do users have easy and constant access to computing?
- Is the task/job that requires support complex enough to warrant an EPSS?
- How stable is the task/job? (if it changes often, an EPSS may be more appropriate as it is easier to maintain, update and disseminate)
- How critical is the task/job? What is the cost of non-compliance or poor performance?
- Is time available for support? (some tasks are so time critical that consulting an EPSS is not realistic)
- Do users possess the necessary characteristics to use an EPSS, in terms of computer literacy, computer expertise or motivation?
- Is turnover among potential users of the EPSS high? (high turnover may justify EPSS as it is more accessible than many other forms of learning)

- Are the logistics of getting people to other alternatives, such as a classroom course, so complicated and expensive that EPSS is a more efficient approach?
- Will the EPSS be used for empowering users to support their compliance with performance standards?
- Is the task repeated frequently? (if not, EPSS may be a sound alternative)
- Are complex decisions involved in the task?
- Can an EPSS be supported/maintained? How will it be updated?

Appendix C - Legal Implications for Learner Verification

The implications for Learner Verification become increasingly apparent with the institution of a performance assessment. As learners are put into the visible and experiential situation of measurement, it becomes more important to provide a fair and positive environment.

Legal issues become correspondingly more apparent in this situation, as the perception of personal risk increases among learners. The user who feels his/her job, rank, or performance evaluation may be affected by the outcomes of the measurement or Learner Verification process may want to challenge the evaluation program. At the very least, morale may suffer.

To eliminate potential user discontent or the raising of legal issues, the <Company Long Name> may want to follow recommended practices. The major points are summarized below.

Areas to Address

Areas to address in creating a legally-defensible testing program

- test validity evidence
- test reliability evidence
- opportunity-to-learn evidence
- adequate test-taker notification
- avoidance-of-bias evidence
- cut score setting
- management of all contractors in testing process
- use of specialists to monitor test development
- provisions for the test retake and grievance processes
- test retake remediation documentation

Stakeholders

Stakeholders must be identified and involved in the process throughout. Stakeholders include the following:

- test-takers
- employers (managers and supervisors)
- test developers and users (managers, administration)
- HRD/legal
- instructional designers and instructors

Communication

Communication of the assessment program needs to start early in the learning cycle and continue through the follow-up phase. The communication model includes

- purpose of assessment
- forms of assessment
- delivery methods
- time frames
- policies for taking tests
- policies concerning re-taking of tests
- opportunities for extended learning

Legal Concepts

Legal issues center on the following concepts, among which there is some overlap

- *Due Diligence* should be exercised to reduce or eliminate errors in measurement resulting in inaccurate classification of individuals
- *Fairness* refers to the absence of bias, the presence of validity, and the equal protection of all groups of test-takers
- *Validation* of test items will make sure of the legal defensibility of the assessments
- *Disparate Treatment* and *Disparate Impact* refer to the equal protection clause. Disparate treatment involved overt illegal discrimination; disparate impact involve employment practices that appear to be neutral, but in fact affect two groups differently. In the event of a legal challenge, plaintiffs have the responsibility to show disparate impact, except in the case of minorities, where the burden of proof shifts to defendants, who are required to demonstrate that tests are job-related and professionally-developed.
- *Adverse* impact refers to a negative impact on the test-takers job or position, based on test results.
- *ADA* (Americans with Disabilities Act) concerns encompass physical considerations as well as diagnosed learning disabilities (for example, Dyslexia)
- *EEOC* (Equal Employment Opportunity Commission) requires that all employees be given equal access to learning opportunities